

Esercizio 1. (Alessandro Martina) Show that in general there is no type $p(x; z) \subseteq L(M)$ such that for every $a \in \mathcal{U}^x$ and $b \in \mathcal{U}^z$

$$a; b \models p(x; z) \Leftrightarrow a \perp_M b.$$

Hint: consider the theory of dense linear orders.

Esercizio 2. (Alessandro Martina) Let M be an arbitrary model. Let $\varphi(x, y) \in L(M)$, where $|x| = |y|$. Prove that there is a sequence $\langle a_i : i < \omega \rangle$ in M^x such that

$$\varphi(a_i, a_j) \leftrightarrow \varphi(a_h, a_k)$$

for every $i < j < \omega$ and $h < k < \omega$.

Esercizio 3. (Francesco Sulpizi) Let $\bar{c} = \langle c_i : i \in I \rangle$ be an A -indiscernible sequence. Let $a \models p(x)$, where $p(x) \in S(A, \bar{c})$ is finitely satisfiable in every $M \supseteq A$. Prove that $\bar{c}$ is indiscernible over A, a .

Esercizio 4. (Francesco Sulpizi) Show that in general there is no type $p(x; z) \subseteq L(M)$ such that for every $a \in \mathcal{U}^x$ and $b \in \mathcal{U}^z$

$$a; b \models p(x; z) \Leftrightarrow a \perp_M b.$$

Hint: consider the theory of dense linear orders.

Esercizio 5. (Leonardo Centazzo) Let $\varphi(x, y) \in L$, where $|x| = |y| = 1$. Suppose there is an infinite set $A \subseteq \mathcal{U}$ such that $\varphi(a, b) \leftrightarrow \varphi(b, a)$ for every two distinct $a, b \in A$. Prove that $\varphi(x; y)$ is unstable.

Esercizio 6. (Leonardo Centazzo) Let $N \geq M$ be λ -saturated for $\lambda > |M|$. Let $a_i \perp_M N, b$, for $i = 1, 2$. Prove that if $a_1 \equiv_N a_2$ then $a_1 \equiv_{N, b} a_2$.

Esercizio 7. (Roberto Carnevale) Let M be a graph with the property that for every finite $A \subseteq M$ there is a $c \in M$ such that $A \subseteq r(c, M)$. The edges of M are colored with infinitely many colors but every clique contains at most k colors, for some fixed $k \in \omega$. Prove that M has an infinite monochromatic complete subgraph.

Esercizio 8. (Roberto Carnevale) Show that in general there is no type $p(x; z) \subseteq L(M)$ such that for every $a \in \mathcal{U}^x$ and $b \in \mathcal{U}^z$

$$a; b \models p(x; z) \Leftrightarrow a \perp_M b.$$

Hint: consider the theory of dense linear orders.

Esercizio 9. (Matteo Bisi) Let M be an arbitrary model. Let $\varphi(x, y) \in L(M)$, where $|x| = |y|$. Prove that there is a sequence $\langle a_i : i < \omega \rangle$ in M^x such that

$$\varphi(a_i, a_j) \leftrightarrow \varphi(a_h, a_k)$$

for every $i < j < \omega$ and $h < k < \omega$.

Esercizio 10. (Matteo Bisi) Let $\bar{c} = \langle c_i : i \in I \rangle$ be an A -indiscernible sequence. Let $a \models p(x)$, where $p(x) \in S(A, \bar{c})$ is finitely satisfiable in every $M \supseteq A$. Prove that $\bar{c}$ is indiscernible over A, a .

Esercizio 11. (Davide Peccioli) Assume L is countable and let T be strongly minimal. Prove that the following are equivalent

1. T is ω -categorical
2. the algebraic closure of a finite set is finite.

Esercizio 12. (Davide Peccioli) Let $\varphi(x, y) \in L$, where $|x| = |y| = 1$. Suppose there is an infinite set $A \subseteq \mathcal{U}$ such that $\varphi(a, b) \leftrightarrow \varphi(b, a)$ for every two distinct $a, b \in A$. Prove that $\varphi(x; y)$ is unstable.

Esercizio 13. (Pietro Giura) Let $N \geq M$ be λ -saturated for $\lambda > |M|$. Let $a_i \perp_M N, b$, for $i = 1, 2$. Prove that if $a_1 \equiv_N a_2$ then $a_1 \equiv_{N, b} a_2$.

Esercizio 14. (Pietro Giura) Let M be a graph with the property that for every finite $A \subseteq M$ there is a $c \in M$ such that $A \subseteq r(c, \mathcal{U})$. The edges of M are colored with infinitely many colors but every clique contains at most k colors, for some fixed $k \in \omega$. Prove that M has an infinite monochromatic complete subgraph.

Esercizio 15. (Costanza Furone) Let $A \subseteq B$ and $p(x) \subseteq L(A)$. Suppose that $\text{tp}(a/B)$ is isolated over B for every $a \models p(x)$. Prove that $p(x)$ is isolated over A .

Esercizio 16. (Costanza Furone) Let M be an arbitrary model. Let $\varphi(x, y) \in L(M)$, where $|x| = |y|$. Prove that there is a sequence $\langle a_i : i < \omega \rangle$ in M^x such that

$$\varphi(a_i, a_j) \leftrightarrow \varphi(a_h, a_k)$$

for every $i < j < \omega$ and $h < k < \omega$.